

SAIKUMAR KALERU

AI ENGINEER (GENAI / LLM APPLICATIONS)

Hyderabad, India | +91 9398320218 | saikumarkaleru56@gmail.com | linkedin.com/in/saikumar-kaleru

PROFESSIONAL SUMMARY

Software engineer with a B.E. in Computer Science and strong Python skills, focused on building applications with large language models. Built a retrieval augmented generation system and a tool using LLM agent from scratch, both behind a clean provider interface that runs on OpenAI or Anthropic and offline for testing. Comfortable with retrieval, embeddings, prompt design, agents and tool calling, and serving models behind REST APIs. Spent close to two years at D. E. Shaw building Python systems and data pipelines over large market datasets. Top 5% in GATE CSE 2021.

TECHNICAL SKILLS

GenAI & LLMs: Retrieval Augmented Generation (RAG), embeddings and vector search, prompt engineering, LLM agents and tool calling (ReAct), structured output, OpenAI and Anthropic APIs

Programming: Python, SQL, C++, data structures and algorithms

Backend & Serving: FastAPI, REST APIs, Docker, provider abstraction, testing (pytest, unittest)

Data: pandas, NumPy, text processing, TF-IDF, chunking, data pipelines

Tools: Git, Linux, CMake, Power BI

EXPERIENCE

D. E. Shaw India Pvt. Ltd. | Member Technical

May 2022 to Jan 2024

Software Engineering, F&O Trading Desk

- Built Python systems and automated data pipelines to capture and process large market datasets.
- Built analytics tools and daily reports that turned raw data into numbers the desk acted on.
- Wrote C++ and Python code in a fast production setting, with unit tests and profiling.
- Cleaned, validated, and reshaped data for analysis, the same groundwork that feeds ML and LLM systems.

Arcesium India Pvt. Ltd. | Site Reliability Engineer (Intern)

Jan 2022 to May 2022

Backend Reliability and Databases

- Managed MS SQL and PostgreSQL databases and improved slow queries.
- Kept AWS EC2 and RDS services available and fixed production issues.

PROJECTS

RAG Document Question Answering | Python, FastAPI

- Built a retrieval augmented generation system with document chunking, TF-IDF and embedding retrieval, and grounded prompts that answer with cited sources.
- Wrote a provider interface for OpenAI, Anthropic, or an offline mock, served it over a CLI and a FastAPI API, and covered it with 25 unit tests.

LLM Agent Toolkit | Python

- Built a ReAct style agent that plans and calls tools (calculator, document search, text and date tools) with a step limit and error handling.
- Added a mock LLM so the agent and its 30 tests run offline, plus OpenAI and Anthropic backends for real use.

REST API Service | Python, FastAPI, SQLAlchemy, Docker

- Built a REST API with token based login, a relational database, and pagination, packaged with Docker.
- Wrote a pytest suite, the kind of service used to serve a model or an LLM feature in production.

EDUCATION

B.E. Computer Science & Engineering, CBIT, Hyderabad

2018 to 2022

Chaitanya Bharathi Institute of Technology · CGPA 8.8

MBA in Finance, UCCBM, Hyderabad

2024 to 2026

University College of Commerce & Business Management · CGPA 8.9

RELEVANT COURSEWORK

Artificial Intelligence, Machine Learning, Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Probability & Statistics, Software Engineering

CERTIFICATIONS

HackerRank: Problem Solving and Java (Basic) Programming

NISM Series-XV: Research Analyst (SEBI/NISM), Reg. No. NISM-202600123995

ACHIEVEMENTS

- GATE CSE 2021: top 5% among national candidates in Computer Science.
- UGC NET 2025 (Management): 95.6 percentile.

ADDITIONAL INFORMATION

Languages: English, Hindi, Telugu · Location: Hyderabad, India (open to relocation) · Availability: Immediate